

EMF & FARADAY'S LAW

Transformer EMF *stationary loop, B changes*

$$V_{\text{emf}} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$$

Motional EMF *loop moves, B static*

$$V_{\text{emf}} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$$

General (flux form) *always works*

$$V_{\text{emf}} = - \frac{d\Phi}{dt} \quad \Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$$

N-turn coil *transformer, N turns*

$$V_{\text{emf}} = -N \frac{d\Phi}{dt} = - \frac{d\lambda}{dt} \quad \lambda = N\Phi$$

Uniform B shortcut *B same everywhere on loop*

$$V_{\text{emf}} = - \frac{\partial B}{\partial t} \cdot A$$

Long wire + rect loop *wire near loop, r: a → b, height c*

$$V_{\text{emf}} = - \int_0^c \int_a^b \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} = - \int_0^c \int_a^b \frac{\partial}{\partial t} \left(\frac{\mu_0 I}{2\pi r} \right) dr dz$$

MAXWELL'S EQUATIONS

Name	Differential form
Faraday	$\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$
Ampère	$\nabla \times \mathbf{H} = \mathbf{J} + \partial \mathbf{D} / \partial t$
Gauss (E)	$\nabla \cdot \mathbf{D} = \rho_v$
Gauss (B)	$\nabla \cdot \mathbf{B} = 0$
$\mathbf{D} = \epsilon \mathbf{E}; \mathbf{B} = \mu \mathbf{H}; \text{source-free: } \rho_v = 0, \mathbf{J} = 0$	

SURFACE ELEMENT *ds*

$d\mathbf{s} = \hat{n} da$ (RHR gives $\hat{n}$: curl fingers in loop direction, thumb = $\hat{n}$)

Loop plane	<i>ds</i>
x-y	$\hat{z} dx dy$ or $\hat{z} r dr d\phi$
y-z	$\hat{x} dy dz$
x-z	$\hat{y} dx dz$
coplanar / toroid	$\hat{\phi} dr dz$

Uniform B: $\iint da = A$ (just the loop area)

DOT PRODUCTS $\hat{a} \cdot \hat{b}$

Same direction = 1; perpendicular = 0

	$\hat{x}$	$\hat{y}$	$\hat{z}$	$\hat{r}$	$\hat{\phi}$
$\hat{x}$	1	0	0	0	0
$\hat{y}$	0	1	0	0	0
$\hat{z}$	0	0	1	0	0
$\hat{\phi}$	0	0	0	0	1

CROSS PRODUCT $\mathbf{A} \times \mathbf{B}$ (determinant)

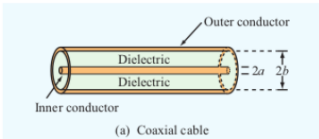
$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = (A_y B_z - A_z B_y) \hat{x} - (A_x B_z - A_z B_x) \hat{y} + (A_x B_y - A_y B_x) \hat{z}$$

Also used for curl:

$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ E_x & E_y & E_z \end{vmatrix}$$

Plane wave ($\partial_x = \partial_y = 0$): only ∂_z row survives.

COAXIAL CABLE — AC RESISTANCE



(a) Coaxial cable

$$R' = \frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right) \quad (\Omega/\text{m})$$

a: inner radius; *b*: outer radius; $R_s = \sqrt{\pi f \mu / \sigma}$

FIELDS

E — electric field (V/m)
D = $\epsilon \mathbf{E}$ — elec. flux density (C/m²)
B — mag. flux density (T)
H — mag. field intensity (A/m)
 ρ_v — charge density (C/m³)
 ρ_{v0} — initial charge density
 $\Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$ — flux (Wb)
 $\lambda = N\Phi$ — flux linkage (Wb)

CURRENTS

$J_c = \sigma E$ — cond. density (A/m²)
 $J_d = \epsilon \partial E / \partial t$ — disp. density
 $I_c = J_c \cdot A$ — cond. current (A)
 $I_d = J_d \cdot A$ — disp. current (A)
 $R = d / (\sigma A)$ — resistance (Ω)
 $C = \epsilon_r \epsilon_0 A / d$ — capacitance (F)
 $\tau_r = \epsilon_r \epsilon_0 / \sigma$ — relax. time (s)

MATERIALS

$\mu_0 = 4\pi \times 10^{-7}$ H/m
 μ_r — relative permeability
 $\mu = \mu_r \mu_0$ — permeability
 $\epsilon_0 = 8.85 \times 10^{-12}$ F/m
 ϵ_r — relative permittivity
 $\epsilon = \epsilon_r \epsilon_0$ — permittivity
 σ — conductivity (S/m)

CIRCUIT & WAVE

V — voltage (V)
R — resistance (Ω)
C — capacitance (F)
 $\omega = 2\pi f$ — angular freq (rad/s)
 $k = \omega \sqrt{\mu \epsilon}$ — wave number (m⁻¹)
 $\lambda = 2\pi / k$ — wavelength (m)
 $c = 3 \times 10^8$ m/s — speed of light

PHASOR & GEOMETRY

$\eta = \sqrt{\mu / \epsilon}$ — intrinsic imp. (Ω)
 $\eta_0 = 120\pi \approx 377 \Omega$ (free space)
 $k / (\omega \mu) = 1 / \eta$
j — imaginary unit ($j^2 = -1$)
E — phasor (drop $e^{j\omega t}$)
A — area (m²); *d* — sep. (m)
N — turns; *r* — radial dist. (m)

B FIELDS

Infinite wire *single wire carrying I, N = 1*

$$\mathbf{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r}$$

Inside toroid *coil of N turns carrying I*

$$\mathbf{B} = \hat{\phi} \frac{\mu N I}{2\pi r} \quad \mu = \mu_r \mu_0$$

N in B only if the source (current-carrying thing) is a multi-turn coil. Single wire ⇒ N = 1, drop it. N in V_{emf} = -NdΦ/dt is the receiver coil turns.

∂B/∂t TABLE

<i>B</i> (<i>t</i>)	$\partial B / \partial t$
$B_0 \sin(\omega t)$	$B_0 \omega \cos(\omega t)$
$B_0 \cos(\omega t)$	$-B_0 \omega \sin(\omega t)$
$B_0 e^{-\alpha t}$	$-\alpha B_0 e^{-\alpha t}$
B_0 (const)	0

SIGN → CURRENT DIRECTION

<i>V</i> _{emf}	Inside source	<i>I</i> dir.
< 0	− → + terminal	$-\hat{\phi}$ (CW)
> 0	+ → − terminal	$+\hat{\phi}$ (CCW)
= 0	—	none

Lenz: induced I always opposes the change in Φ.

ln INTEGRAL

B ∝ 1/r, loop spans r = a to r = b

$$\int_a^b \frac{dr}{r} = \ln \frac{b}{a} = \ln b - \ln a$$

$$\ln \frac{b}{a} = \ln b - \ln a$$

$$\ln 2 = 0.693 \quad \ln 3 = 1.099 \quad \ln 1.2 = 0.182$$

ROTATING LOOP — GENERATOR

$\Phi = AB_0 \cos(\omega t + \phi_0)$ $V_{\text{emf}} = AB_0 \omega \sin(\omega t + \phi_0)$ $I_{\text{amp}} = \frac{AB_0 \omega}{R} \quad \omega = 2\pi f = 2\pi \frac{\text{rpm}}{60}$ Common loop areas	
Shape	Area <i>A</i>
Square (side <i>a</i>)	a^2
Rectangle (<i>l</i> × <i>w</i>)	lw
Circle (radius <i>r</i>)	πr^2
Equil. triangle (side <i>a</i>)	$\frac{\sqrt{3}}{4} a^2$
Any triangle	$\frac{1}{2} bh$

SI PREFIXES

Prefix	Sym	Value	Prefix	Sym	Value
pico	p	10 ⁻¹²	kilo	k	10 ³
nano	n	10 ⁻⁹	mega	M	10 ⁶
micro	μ	10 ⁻⁶	giga	G	10 ⁹
milli	m	10 ⁻³	tera	T	10 ¹²

j IDENTITIES

Expression	Result	Expression	Result
j^2	−1	$j \times (-j)$	1
$1/j$	− <i>j</i>	$-1/j$	<i>j</i>
j^3	− <i>j</i>	j^4	1
$e^{-j\pi/2}$	− <i>j</i>	$e^{j\pi/2}$	<i>j</i>
$e^{j\pi}$	−1	$e^{j2\pi}$	1

IDEAL TRANSFORMER

Primary: *N*₁ turns, *V*₁, *I*₁; Secondary: *N*₂ turns, *V*₂, *I*₂, load *R*_L

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} \quad \frac{I_1}{I_2} = \frac{N_2}{N_1}$$
$$R_{\text{in}} = \left(\frac{N_1}{N_2} \right)^2 R_L \quad Z_{\text{in}} = \left(\frac{N_1}{N_2} \right)^2 Z_L$$

DISPLACEMENT & CONDUCTION CURRENT

Conduction *resistive, σ ≠ 0*

$$E = \frac{V(t)}{d} \quad J_c = \sigma E \quad I_c = J_c \cdot A = \frac{\sigma A}{d} V(t)$$

$$I_c = \frac{\sigma A}{d} V(t) \longleftrightarrow I = \frac{V}{R} \Rightarrow R = \frac{d}{\sigma A}$$

$1/R$

Current proportional to *V* directly → behaves like a resistor.

Displacement *capacitive, changing E*

$$J_d = \epsilon \frac{\partial E}{\partial t} \quad I_d = J_d \cdot A = \frac{\epsilon_r \epsilon_0 A}{d} \frac{dV}{dt}$$

$$I_d = \underbrace{\frac{\epsilon_r \epsilon_0 A}{d}}_C \frac{dV}{dt} \longleftrightarrow I = C \frac{dV}{dt} \Rightarrow C = \frac{\epsilon_r \epsilon_0 A}{d}$$

Current proportional to *dV/dt* → behaves like a capacitor.

Lossy cap circuit *both σ and ε_r present*

$$R \parallel C \text{ across } V(t): I = I_c + I_d = \frac{V}{R} + C \frac{dV}{dt}$$

CHARGE-CURRENT CONTINUITY

$$\nabla \cdot \mathbf{J} = - \frac{\partial \rho_v}{\partial t}$$

Conductor (steady): $\nabla \cdot \mathbf{J} = 0 \Rightarrow \oint_S \mathbf{J} \cdot d\mathbf{s} = 0$ (KCL)

CHARGE RELAXATION

Decay *charge injected at t = 0*

$$\rho_v(t) = \rho_{v0} e^{-t/\tau_r} \quad \tau_r = \frac{\epsilon_r \epsilon_0}{\sigma}$$

Find σ from decay data

$$\sigma = \frac{-\ln(\rho_v / \rho_{v0}) \cdot \epsilon_r \epsilon_0}{t_1}$$

Larger τ_r = slower dissipation = better insulator.

PHASOR METHOD (E → H)

1. Key relation $\mathbf{E}(z, t) = \text{Re}\{\tilde{\mathbf{E}}(z) e^{j\omega t}\}$
2. Euler split $e^{j(\omega t - kz)} = e^{j\omega t} \cdot e^{-jkz}$ strip $e^{j\omega t}$, keep e^{-jkz}

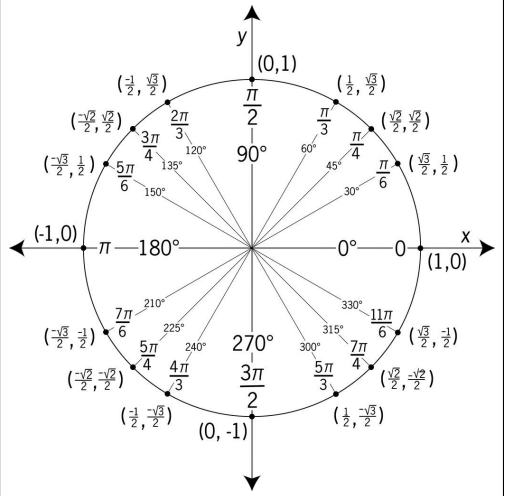
3. Time → phasor conversions
 $\cos(\omega t - kz) = \text{Re}\{e^{j(\omega t - kz)}\} = \text{Re}\{e^{j\omega t} \cdot e^{-jkz}\} \rightarrow e^{-jkz}$
 $\sin(\omega t - kz) = \cos(\omega t - kz - \frac{\pi}{2}) = \text{Re}\{e^{j(\omega t - kz - \pi/2)}\} = \text{Re}\{e^{-j\pi/2} \cdot e^{j\omega t} \cdot e^{-jkz}\} \rightarrow -j e^{-jkz}$

4. Faraday in phasor domain (plane wave: only $\partial/\partial z$ survives)

$$\tilde{\mathbf{H}} = - \frac{1}{j\omega \mu} \nabla \times \tilde{\mathbf{E}}$$

5. Phasor → time domain $\mathbf{H}(z, t) = \text{Re}\{\tilde{\mathbf{H}} e^{j\omega t}\}$
 $\text{Re}\{e^{j(\omega t - kz)}\} = \cos(\omega t - kz)$
 $\text{Re}\{-j e^{j(\omega t - kz)}\} = \sin(\omega t - kz)$

UNIT CIRCLE



PLANE WAVE PARAMETERS (LOSSLESS)

Angular frequency ω : rad/s; f : Hz
 $\omega = 2\pi f \quad f = \frac{\omega}{2\pi}$

Phase velocity m/s ; *nonmagnetic*: $\mu_r = 1$
 $u_p = \frac{c}{\sqrt{\epsilon_r}} \quad c = 3 \times 10^8 \text{ m/s}$

Wavenumber k rad/m
 $k = \frac{\omega}{u_p} = \frac{2\pi}{\lambda} = \frac{\omega\sqrt{\epsilon_r}}{c}$

Wavelength & impedance $m \text{ \& } \Omega$
 $\lambda = \frac{u_p}{f} = \frac{2\pi}{k} \text{ (m)} \quad \eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)}, \quad \eta_0 = 377 \Omega$

Order: $\omega \rightarrow u_p \rightarrow k \rightarrow \lambda \rightarrow \eta$. Base: $\mathbf{E} = E_0 \cos(\omega t \pm kx + \phi_0)\hat{e}$.

LOSSLESS PLANE WAVE — E AND H

General E $\mathbf{E} = E_0 \cos(\omega t - k\hat{k} \cdot \mathbf{r} + \phi_0)\hat{e}$
 $\hat{k}$ = direction of travel (unit vector), e.g. $+y \Rightarrow \hat{k} = \hat{y}$.
 k = wavenumber (rad/m): how fast phase changes in space;
 $k = \omega\sqrt{\epsilon_r}/c = 2\pi/\lambda$.
 ϕ_0 = initial phase constant (rad): shift set by given condition at known t , position.
 $\hat{e}$ = polarization direction of $\mathbf{E}$ (unit vector), e.g. E along $x \Rightarrow \hat{e} = \hat{x}$.
 $\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$, so $\hat{k} \cdot \mathbf{r}$ picks the axis, e.g. $\hat{k} = \hat{y} \Rightarrow \hat{k} \cdot \mathbf{r} = y$.
 $\mathbf{H}$ from $\mathbf{E}$, $\hat{k}$, $\hat{e}$, $\hat{H}$ form right-hand set
 $\mathbf{H} = \frac{1}{\eta} (\hat{k} \times \mathbf{E})$

Find ϕ_0 given $E = E_1$ at $t = 0$, position r_0
 $E_0 \cos(-kr_0 + \phi_0) = E_1 \Rightarrow \phi_0 = \cos^{-1}\left(\frac{E_1}{E_0}\right) + kr_0$
 $\cos^{-1}$ gives $[0, \pi]$; take ϕ_0 in that range.

LOSSLESS vs. LOSSY — E & H

	Lossless ($\sigma = 0$)	Lossy ($\sigma \neq 0$)
Impedance	$\eta = \sqrt{\mu/\epsilon}$	$\eta_c = \eta_c e^{j\theta_\eta}$
Prop.	e^{-jkz}	$e^{-\alpha z} e^{-j\beta z}$
$\mathbf{E}(t)$	$\text{Re}\{\tilde{\mathbf{E}} e^{j\omega t}\}$ (same)	
$\mathbf{H}(t)$	$\text{Re}\{\tilde{\mathbf{H}} e^{j\omega t}\}$ (same)	
$\tilde{\mathbf{H}}$ from $\tilde{\mathbf{E}}$	$\frac{1}{\eta} (\hat{k} \times \tilde{\mathbf{E}})$	$\frac{1}{\eta_c} (\hat{k} \times \tilde{\mathbf{E}})$
$\tilde{\mathbf{E}}$ from $\tilde{\mathbf{H}}$	$-\eta (\hat{k} \times \tilde{\mathbf{H}})$	$-\eta_c (\hat{k} \times \tilde{\mathbf{H}})$
Phase	in phase	$\mathbf{H}$ lags $\mathbf{E}$ by θ_η
Lossless shortcuts (wave in $+z$, $\mathbf{E}$ along $\hat{e}$):		
$\mathbf{E}(z, t)$	$E_0 \cos(\omega t - kz + \phi) \hat{e}$	
$\mathbf{H}(z, t)$	$\frac{E_0}{\eta} \cos(\omega t - kz + \phi) (\hat{k} \times \hat{e})$	
$\mathbf{H}(t)$ direct	$\frac{1}{\eta} (\hat{k} \times \mathbf{E}(t))$	
$\mathbf{E}(t)$ direct	$-\eta (\hat{k} \times \mathbf{H}(t))$	

DIRECTION OF PROPAGATION

Arg. contains	$\hat{k}$	Arg. contains	$\hat{k}$
$\omega t - kz$	$+\hat{z}$	$\omega t + kz$	$-\hat{z}$
$\omega t - ky$	$+\hat{y}$	$\omega t + ky$	$-\hat{y}$
$\omega t - kx$	$+\hat{x}$	$\omega t + kx$	$-\hat{x}$

Minus before k -axis $\Rightarrow +$ direction. Plus $\Rightarrow -$ direction.

$\hat{k} \times \hat{E}$ — FIND $\hat{H}$ DIRECTION

$\hat{k}$	$\hat{E}$	$\hat{H}$	$\hat{k}$	$\hat{E}$	$\hat{H}$
$+\hat{z}$	$\hat{x}$	$+\hat{y}$	$+\hat{z}$	$\hat{y}$	$-\hat{x}$
$-\hat{z}$	$\hat{x}$	$-\hat{y}$	$-\hat{z}$	$\hat{y}$	$+\hat{x}$
$+\hat{y}$	$\hat{x}$	$-\hat{z}$	$+\hat{y}$	$\hat{z}$	$+\hat{x}$
$-\hat{y}$	$\hat{x}$	$+\hat{z}$	$-\hat{y}$	$\hat{z}$	$-\hat{x}$
$+\hat{x}$	$\hat{y}$	$+\hat{z}$	$+\hat{x}$	$\hat{z}$	$-\hat{y}$
$-\hat{x}$	$\hat{y}$	$-\hat{z}$	$-\hat{x}$	$\hat{z}$	$+\hat{y}$

Neg. $\hat{E}$: flip $\hat{H}$ sign. $\hat{k} \times \hat{k} = \mathbf{0}$; any $\hat{a} \times \hat{a} = \mathbf{0}$.
 Cyclic: $\hat{x} \times \hat{y} = \hat{z}$, $\hat{y} \times \hat{z} = \hat{x}$, $\hat{z} \times \hat{x} = \hat{y}$. Reverse $\Rightarrow$ flip sign.

TRIG PHASE SHIFTS ($\theta \equiv \omega t - kz$)

Input	Result	Input	Result
$\sin(\theta + \pi/2)$	$\cos \theta$	$\cos(\theta + \pi/2)$	$-\sin \theta$
$\sin(\theta - \pi/2)$	$-\cos \theta$	$\cos(\theta - \pi/2)$	$\sin \theta$
$\sin(\theta + \pi)$	$-\sin \theta$	$\cos(\theta + \pi)$	$-\cos \theta$
$\sin(\theta - \pi)$	$-\sin \theta$	$\cos(\theta - \pi)$	$-\cos \theta$

WAVE PARAMS

$\omega = 2\pi f$ — ang. freq (rad/s)
 $k = \omega\sqrt{\epsilon_r}/c$ — wavenumber (rad/m)
 $\lambda = 2\pi/k$ — wavelength (m)
 $u_p = c/\sqrt{\epsilon_r}$ — phase vel. (m/s)
 $\eta = \eta_0/\sqrt{\epsilon_r}$ — impedance (Ω)
 $c = 3 \times 10^8 \text{ m/s}$
 $\eta_0 = 377 \Omega$ (free space)

LOSSLESS MEDIA

$\sigma = 0$ (vacuum, air, ideal dielectric)
 $\epsilon'' = \sigma/\omega = 0 \Rightarrow \epsilon = \epsilon'$ (real)
 $\alpha = 0$; no attenuation
 $\eta_c = \eta_0/\sqrt{\epsilon_r}$ real (Ω)
 $\eta_0 = 377 \Omega$ (free space)
 $\mathbf{E}$ and $\mathbf{H}$ in phase ($\theta_\eta = 0$)

LOSSY MEDIA

α — atten. const (Np/m)
 β — phase const (rad/m)
 $\gamma = \alpha + j\beta$ — prop. const (m $^{-1}$)
 η_c — complex impedance (Ω)
 $\epsilon'' = \epsilon_r \epsilon_0$; $\epsilon'' = \sigma/\omega$ (F/m)
 μ_r — rel. permeability ($\mu_r = 1$: nonmagnetic)
 $\mu = \mu_r \mu_0$; $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$
 $\delta_s = 1/\alpha$ — skin depth (m)

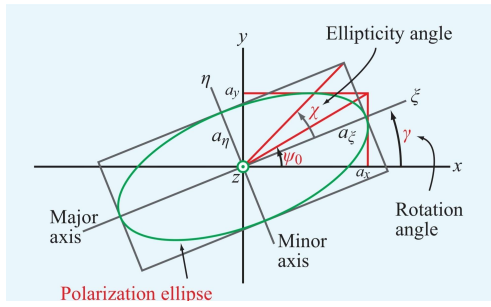
CIRCULAR WAVES

$\tilde{\mathbf{E}}_R = a_R(\hat{x} + e^{-j\pi/2}\hat{y})e^{-jkz} = a_R(\hat{x} - j\hat{y})e^{-jkz}$
 $\tilde{\mathbf{E}}_L = a_L(\hat{x} + e^{+j\pi/2}\hat{y})e^{-jkz} = a_L(\hat{x} + j\hat{y})e^{-jkz}$
 a_R, a_L, a_x in V/m; k in rad/m
 Linear: $a_R = a_L = a_x/2$
 $-j = e^{-j\pi/2}$: $\cos \rightarrow \sin$
 $+j = e^{+j\pi/2}$: $\cos \rightarrow -\sin$

USEFUL NUMBERS

$\sqrt{2.56} = 1.6$; $\sqrt{9} = 3$
 $\sqrt{2.5} \approx 1.581$; $\sqrt{2} \approx 1.414$
 $\arccos(2/3) \approx 48.2^\circ = 0.84 \text{ rad}$
 $\arctan(4/3) \approx 53.1^\circ = 0.927 \text{ rad}$
 $\arctan(3/4) \approx 36.9^\circ = 0.644 \text{ rad}$
 $\arctan(24/7) \approx 73.7^\circ = 1.287 \text{ rad}$

POLARIZATION — GENERAL WAVE



$\mathbf{E} = a_x \cos(\omega t - kz)\hat{x} + a_y \cos(\omega t - kz + \delta)\hat{y}$
 a_x, a_y : bounding box half-widths (V/m); δ : phase diff (rad)
 $\delta = 0$: $\mathbf{E} = (a_x\hat{x} + a_y\hat{y})\cos(\omega t - kz)$ (linear)
 $\delta = \pi$: $\mathbf{E} = (a_x\hat{x} - a_y\hat{y})\cos(\omega t - kz)$ (linear)
 Auxiliary angle $\psi_0 = \arctan(a_y/a_x) \in (0, \pi/2)$
 Rotation angle γ (major axis from $\hat{x}$) $\gamma \in [0, \pi)$
 $\tan(2\gamma) = \tan(2\psi_0) \cos \delta$
 If $\text{sign}(\gamma) \neq \text{sign}(\delta)$: $\gamma \leftarrow \gamma \pm \frac{\pi}{2}$.
 Ellipticity angle χ $\chi \in [-\pi/4, \pi/4]$
 $\sin(2\chi) = \sin(2\psi_0) \sin \delta$
 Axial ratio $R = 1/\tan|\chi|$ major axis = $R \times$ minor; $R \geq 1$

χ SHAPE & HANDEDNESS

χ describes the shape of the ellipse and rotation direction.

χ (rad)	Polarization Type	Handedness
0	Linear	None
$(0, \pi/4)$	RHEP	RH (CCW)
$\pi/4$	RHC	RH (CCW)
$-\pi/4$	LHC	LH (CW)
$(-\pi/4, 0)$	LHEP	LH (CW)

QUADRANT RULE FOR $\tan(2\gamma)$

$N = \sin(2\psi_0) \cos \delta, \quad D = \cos(2\psi_0)$:

N	D	2γ (deg)	2γ (rad)
+	+	$\arctan(N/D)$	$\arctan(N/D)$
+	-	$180^\circ - \arctan N/D $	$\pi - \arctan N/D $
-	-	$180^\circ + \arctan N/D $	$\pi + \arctan N/D $
-	+	$360^\circ - \arctan N/D $	$2\pi - \arctan N/D $

$\gamma = (2\gamma)/2. \quad D = 0 \Rightarrow 2\gamma = \frac{\pi}{2} \Rightarrow \gamma = \frac{\pi}{4}$.
 Convert: $\theta_{\text{rad}} = \theta_0 \times \frac{\pi}{180}$; $\theta_0 = \theta_{\text{rad}} \times \frac{180}{\pi}$.

PHASOR $\leftrightarrow$ TIME DOMAIN

Phasor coeff	$\text{Re}\{(\cdot)e^{j\omega t}e^{-j\beta z}\}$	Result
A	$\text{Re}\{Ae^{j(\omega t - \beta z)}\}$	$A \cos(\omega t - \beta z)$
$-jA$	$\text{Re}\{Ae^{j(\omega t - \beta z - \pi/2)}\}$	$A \sin(\omega t - \beta z)$
$+jA$	$\text{Re}\{Ae^{j(\omega t - \beta z + \pi/2)}\}$	$-A \sin(\omega t - \beta z)$
$-A$	$\text{Re}\{Ae^{j(\omega t - \beta z \pm \pi)}\}$	$-A \cos(\omega t - \beta z)$
$Ae^{j\phi}$	$\text{Re}\{Ae^{j(\omega t - \beta z + \phi)}\}$	$A \cos(\omega t - \beta z + \phi)$
$A + jB$	$\text{Re}\{ Z e^{j(\omega t - \beta z + \theta)}\}$	$ Z \cos(\omega t - \beta z + \theta)$
$ Z = \sqrt{A^2 + B^2}, \quad \theta = \arctan(B/A)$ (check quadrant)		

CIRCULAR POLARIZATION — TIME DOMAIN

$a_x = a_y = a$, wave in $+\hat{z}$
 RHC $\delta = -\pi/2$; $\tilde{\mathbf{E}} = a(\hat{x} - j\hat{y})e^{-jkz}$
 $\mathbf{E}(z, t) = a[\cos(\omega t - kz)\hat{x} + \sin(\omega t - kz)\hat{y}]$
 LHC $\delta = +\pi/2$; $\tilde{\mathbf{E}} = a(\hat{x} + j\hat{y})e^{-jkz}$
 $\mathbf{E}(z, t) = a[\cos(\omega t - kz)\hat{x} - \sin(\omega t - kz)\hat{y}]$

LOSSY MEDIA — CLASSIFICATION

Loss tangent

ϵ''/ϵ'	Type	Use
> 100	Good conductor	GC approx
0.01 to 100	Quasi-conductor	Exact eqs
< 0.01	Low-loss dielectric	LD approx

$\epsilon' = \epsilon_r \epsilon_0$; $\epsilon'' = \sigma/\omega$; nonmag.: $\mu = \mu_0$; free space: $\epsilon_r = \mu_r = 1$.

ANY MEDIUM — EXACT ($X = \sqrt{1 + (\epsilon''/\epsilon')^2}$)

$\alpha = \omega\sqrt{\frac{\mu\epsilon'}{2}}\sqrt{X-1} \text{ Np/m}, \quad \beta = \omega\sqrt{\frac{\mu\epsilon'}{2}}\sqrt{X+1} \text{ rad/m}$
 $\lambda = \frac{2\pi}{\beta} = \frac{u_p}{f} \text{ (m)}, \quad u_p = \frac{\omega}{\beta} \text{ (m/s)}$
 $\eta_c = \frac{\eta}{X^{1/2}} e^{j\theta_\eta} \text{ (}\Omega\text{)}, \quad \theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega\epsilon'} \text{ (rad)}$
 Rect: $\eta_c = |\eta_c| \cos \theta_\eta + j|\eta_c| \sin \theta_\eta, \quad |\eta_c| = \eta/X^{1/2}$
 $\eta = \eta_0/\sqrt{\epsilon_r}$; $\epsilon' = \epsilon_r \epsilon_0$.
 Given $\mathbf{H} \propto e^{-\alpha y} e^{-j\beta y}$: $\gamma = \alpha + j\beta$; $\eta_c = j\omega\mu_0/\gamma = \omega\mu_0(\beta + j\alpha)/(\alpha^2 + \beta^2)$

SKIN DEPTH δ_s (m) — ANY LOSSY MEDIUM

General: $\delta_s = 1/\alpha$ $|E(z)| = E_0 e^{-z/\delta_s}$; lossless: $\alpha = 0 \Rightarrow$ no decay
 Low-loss ($\epsilon''/\epsilon' \ll 1$) $\delta_s \approx 2/(\sigma\eta)$
 Good conductor ($\epsilon''/\epsilon' \gg 1$) $\delta_s \approx 1/\sqrt{\pi f \mu \sigma}$

LOW-LOSS MEDIUM ($\epsilon''/\epsilon' \ll 1$)

$\alpha \approx \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon'}} = \frac{\sigma\eta}{2} \text{ (Np/m)}$
 $\beta \approx \omega\sqrt{\mu\epsilon'} = \frac{\omega\sqrt{\epsilon_r}}{c} \text{ (rad/m)}, \quad u_p \approx \frac{c}{\sqrt{\epsilon_r}} \text{ (m/s)}$
 $\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}} = \eta \text{ (}\Omega\text{)}, \quad \angle \approx 0$

GOOD CONDUCTOR ($\epsilon''/\epsilon' \gg 1$)

$\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma} \text{ (Np/m, rad/m)}$
 $u_p \approx \sqrt{4\pi f / \mu \sigma} \text{ (m/s)}, \quad \lambda \approx u_p/f \text{ (m)}$
 $\eta_c \approx (1 + j)\sqrt{\frac{\pi f \mu}{\sigma}} \approx (1 + j)\frac{\sigma}{\omega} \text{ (}\Omega\text{)}$

SURFACE RESISTANCE (GOOD CONDUCTOR)

(a) Exponentially decaying $\tilde{J}_s(z)$

$J_0 = \sigma E_0$ — current density at surface ($z=0$)
 $\tilde{\mathbf{J}}(z) = J_0 e^{-(1+j)z/\delta_s} \hat{x}$ — decays with depth
 $R_s = \sqrt{\pi f \mu / \sigma} = 1/(\sigma \delta_s) \text{ (}\Omega\text{)}$
 $Z_s = (1 + j)R_s$ — surface impedance
 $R_{AC} = R_s \ell / w$ — AC resistance of slab

AVERAGE POWER DENSITY (POYNTING)

General W/m^2 $\mathbf{S}_{av} = \frac{1}{2} \text{Re}\{\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*\}$
 Lossless shortcut $|\mathbf{E}_0|^2 = a_x^2 + a_y^2 + a_z^2$
 $\mathbf{S}_{av} = \frac{|\mathbf{E}_0|^2}{2\eta} \hat{k}$
 Lossy medium $\eta_c = |\eta_c| e^{j\theta_\eta}$; wave in $+\hat{z}$
 $\mathbf{S}_{av} = \frac{|\tilde{\mathbf{E}}(0)|^2}{2|\eta_c|} e^{-2\alpha z} \cos \theta_\eta \hat{z}$
 Direction = $\hat{k}$. If wave goes in $-\hat{x}$, then $\mathbf{S}_{av}$ is a negative x -component.

DECIBELS & ATTENUATION

$G = 10 \log(P_1/P_2) = 20 \log(|E_1|/|E_2|) \text{ (dB)}$
 $A = -8.69 \alpha z \text{ (dB)}; \quad \alpha \text{ (dB/m)} \equiv 8.69 \alpha \text{ (Np/m)}$